Manuscript: Sample-Size Phase Transition Ends the Hard-versus-Soft Clustering Debate
Hard-versus-Soft Clustering Debate in Causal Discovery

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Suggested Reviewers

We respectfully suggest five reviewers covering causal discovery methodology, clustering theory, computational genomics, and statistical rigor—the four pillars of our contribution. All contact information is publicly available from institutional directories and individual homepages.

1. Caroline Uhler

MIT EECS & Broad Institute of MIT and Harvard

Email: `cuhler@mit.edu`

Prof. Uhler directs the Eric and Wendy Schmidt Center at the Broad Institute and is a leading figure in causal discovery for genomics. Her work spans graphical models, causal inference, and gene regulatory network reconstruction from single-cell and bulk transcriptomic data (e.g., *PLoS Comp. Biol.*, 2026). As a researcher at the intersection of causal discovery methodology and cancer genomics, Prof. Uhler is uniquely qualified to evaluate both our methodological contribution and the biological significance of our pan-cancer findings.

2. Kun Zhang

Carnegie Mellon University / MBZUAI

Email: `kunz1@andrew.cmu.edu`

Prof. Zhang is an internationally recognized causal discovery researcher whose work on ICA-LiNGAM, causal representation learning, and domain adaptation provides an independent perspective on our NOTEARS-based framework. Unlike NOTEARS-lineage researchers, his expertise lies in functional causal models and non-Gaussian methods, enabling an impartial evaluation of whether cluster-aware gating generalizes beyond the continuous-optimization paradigm. His recent work on causal discovery from gene expression data (*arXiv*:2606.00568, 2026) further confirms his relevance to our empirical setting.

3. Marina Meilă

University of Waterloo

Email: `mmp@uwaterloo.ca`

Prof. Meilă holds a Canada Research Chair in Reliable Structure Discovery and is one of the world's foremost authorities on clustering theory. Her foundational work on spectral clustering, clustering consistency, and manifold-based methods makes her the ideal reviewer for a paper whose central claim is resolving a two-decade debate between hard and soft clustering.

4. Fabian J. Theis

Helmholtz Munich & Technical University of Munich

Email: `fabian.theis@helmholtz-munich.de`

Prof. Theis is a leading computational biologist and a core developer of the Scanpy ecosystem for single-cell genomics. His work on single-cell data integration, dimensionality reduction, and deep generative models for transcriptomics bridges methodology and biomedical application. His expertise in single-cell RNA-seq analysis makes him uniquely suited to evaluate our

PBMC replication, cross-modal phase transition evidence, and the practical $n/d \approx 4$ decision rule for genomics applications.

5. Jonas Peters
University of Copenhagen
Email: `jonas.peters@math.ku.dk`

Prof. Peters is a leading causal inference methodologist whose work on Invariant Causal Prediction and distributional robustness provides the statistical rigor needed to evaluate our causal claims. His perspective complements the other four reviewers by assessing whether our empirical findings satisfy the statistical criteria for causal (rather than merely associational) interpretation in observational transcriptomic data.

We confirm that none of the above five individuals have known conflicts of interest with this work. All are independent of the authors and their institutions; none has co-authored with any author within the past three years; and none has served as an advisor or advisee of any author.

Note: All institutional email addresses are publicly verifiable through the respective university directories, individual CVs, and Google Scholar profiles. Should the editor require alternative candidates, we are happy to provide additional names on request.